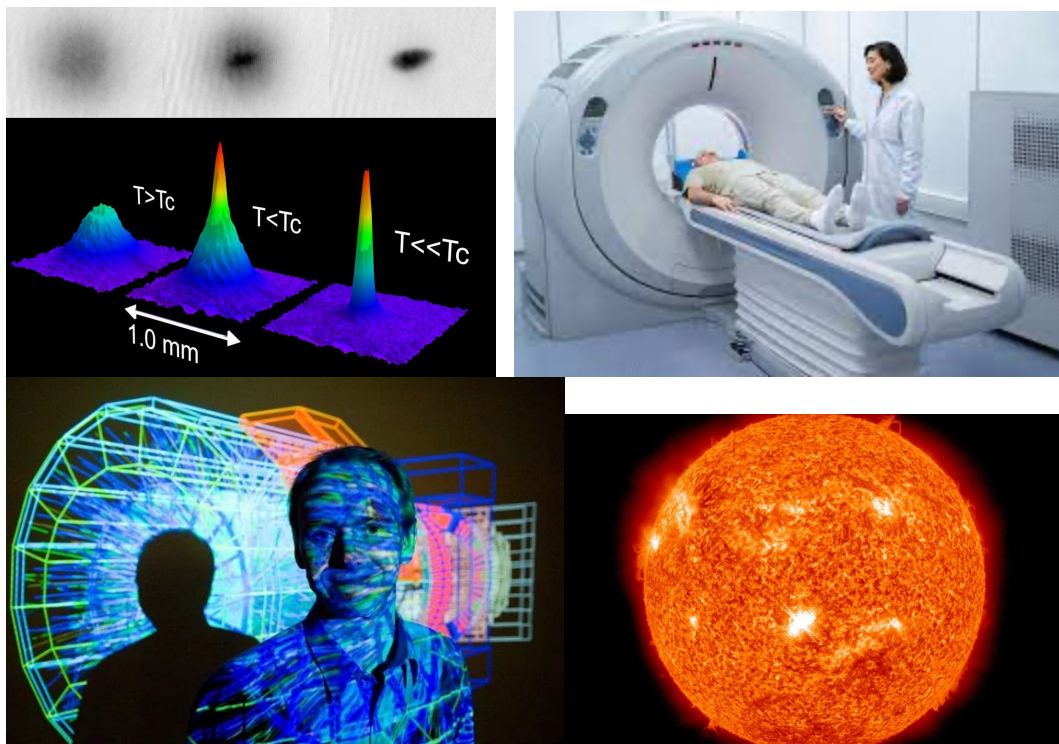


*Thermodynamics is the only physical theory of universal content which I am convinced, that within the framework of applicability of its basic concepts, will never be overthrown.*

– Albert Einstein, Quoted in M Klein "Thermodynamics in Einstein's Thought".

## 1 Introduction



**Figure 1.1:** Some uses of thermal physics. Clockwise from top left: Bose-Einstein condensation of atoms at  $10^{-9}$  K; MRI imaging; the Sun and other stars; the early Universe and the quark gluon plasma at the Alice experiment at CERN. Credits: JPL, NASA; Yashoda healthcare; National Geographic; Birmingham Mail

There are two approaches to describing macroscopic matter (Avogadro's number of particles,  $N_A$ ) at a nonzero temperature.

Consider the air in a lecture theatre. We could specify the volume, the pressure and the temperature, and with greater difficulty, the number of molecules, assuming the doors and windows are shut. On the other hand we could (in principle!) specify the trajectories of the molecules,  $\mathbf{r}_n(t)$ , with  $n = 1, 2, \dots, N_A$ , or perhaps, more readily, specify their motion in a statistical manner.

These mirror the discussion in part of Y1 *Temperature and Matter*: thermodynamics and kinetic theory. We will develop both of these approaches further in this module.

Why are both needed?

Thermodynamics provides general relations between physical quantities – some easy to measure, some not – and input of data to make predictions about other measurements. This is all *without* considering (potentially unknown) microscopic details. We will see many examples of this.

On the other hand, statistical physics, or statistical mechanics, allows prediction of macroscopic measurements from a microscopic basis, providing values for quantities which can only be measured in thermodynamics. It can also provide physical insight – as in the kinetic description of pressure last year.

*Entropy* will be a central quantity in this module, initially as a mysteriously central entity in thermodynamics, especially in the Second Law, and then revealed as a measure of microscopic disorder in statistical physics.

## 1.1 Thermodynamic preliminaries

We start with defining the system we are studying, its boundary and the surroundings (the rest of the Universe) depicted in Fig. (1.2). There we see a prototypical example: the system being gas contained in a piston (constituting the boundary) with the atmosphere outside.



**Figure 1.2:** The system, its boundary and the surroundings. An example being gas in a piston.

The boundary is important in three ways for simple systems:

- (i) whether it allows heat to be exchanged with surroundings: *diathermal* if so, *adiabatic* if not;
- (ii) whether the boundary can move – to allow work to be done on or by the system;

- (iii) whether matter may be exchanged with surroundings, the boundary being *impermeable* if not. In this module we will almost always assume impermeability of boundaries, with the discussion of permeability being in Y3 Statistical Physics.

Thermal *equilibrium* implies at a macroscopic level the system does not vary with time. For a gas this means there are no macroscopic flows in the gas, although its constituents may be moving at a microscopic level. An implication is that the pressure in a gas is spatially uniform – otherwise flows would occur. Equilibrium is more restricted than mechanical equilibrium: most thermal systems are in stable equilibrium. *Metastable* equilibrium also occurs, with timescales from seconds to centuries. When there are no barriers to equilibration, the times to reach equilibrium can be very short: for example  $10^{-12}$  s for electrons in a metal.

Simple thermodynamic systems usually have two variables to define their equilibrium "state". For a gas they could be the pressure,  $p$ , and the volume,  $V$ , the gas occupies. For a magnetic system a natural choice is the strength of an applied field,  $\mathbf{H}$ , and the magnetisation,  $\mathbf{M}$ . For a solid one needs more parameters – the state of strain and stress. There are more complicated systems - liquid crystals and some phases of quantum gases and liquids (and more complicated types of magnetic order).

## 1.2 Zeroth law of thermodynamics – the origin of temperature

Take two systems in equilibrium separately and bring them together, so they can exchange heat. If their states, when in common equilibrium, are defined by  $(p_A, V_A)$  and  $(p_B, V_B)$  then we find experimentally that if one fixes, say,  $p_A, V_A$  and  $V_B$ , then one must vary  $p_B$ , to ensure equilibrium occurs (i.e. the other parameters can remain fixed). Thus there is a relation  $f_{A,B}(p_A, V_A; p_B, V_B) = 0$ .

Now consider three bodies,  $A, B$  and  $C$ ,  $C$  can exchange energy with  $A$  and  $B$  (i.e. the walls between them are diathermal) but are together surrounded by adiabatic walls with respect to the rest of the universe.

If as a result of the diathermal walls  $A$  and  $C$  are at equilibrium with each other, and  $B$  and  $C$  are, then  $A$  and  $B$  are in equilibrium as well. This is the *zeroth law of thermodynamics* and is an experimental fact.

To bring out the consequence of this, note that if  $C$  were a thermometer, then its temperature would be that of  $A$  and  $B$ , so all three bodies are at a common temperature,  $T$ . A common temperature is an expression of bodies being in equilibrium.

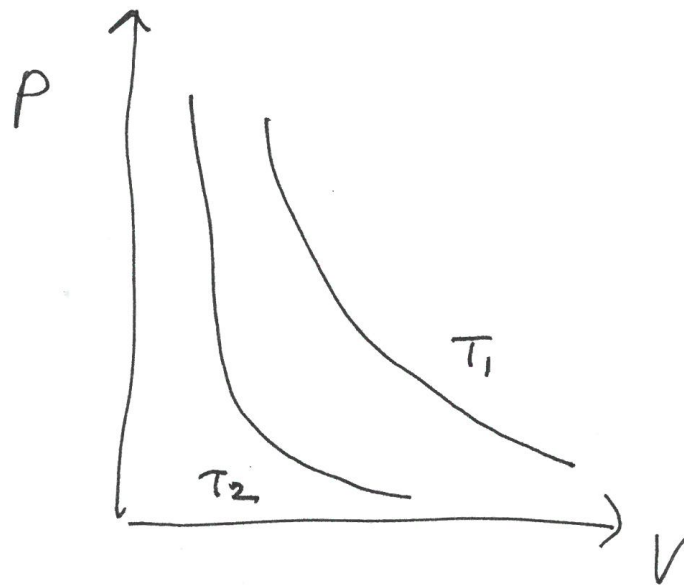
What is less obvious is that this result implies that there is an *equation of state*

$$\phi(p, V) = T . \quad (1.1)$$

(See Appendix.) Although the existence of the equation of state has a thermodynamic origin, its functional form,  $\phi(p, V)$ , be derived within thermodynamics – it either comes from experiment, or from a calculation in statistical mechanics. Note the form of the function  $\phi(p, V)$  depends on the substance – for example  $\text{H}_2\text{O}$  and Ar have very different equations of state if examined near room temperature.

In principle, the equation of state may be solved to obtain  $p = p(V, T)$  which one may then plot in the  $p$ - $V$  plane. The choice of which of  $p$  or  $V$  is naturally the "x-axis" or "y-axis" is determined by which one may vary and measure most easily. Measuring the *volume*, and controlling it (e.g. via a piston), is easier, so making  $V$  the "*independent variable*" and  $p(V)$  the "*dependent variable*" is more natural.

In the Fig. (1.3) we see the *isotherms* for two different temperatures,  $T_1$  and  $T_2$ . To see the



**Figure 1.3:** Representing the isotherms  $p = p(V, T_1)$  and  $p = p(V, T_2)$ .

order of isotherms, note we find experimentally that for fixed  $V$ ,  $p$  increases as  $T$  is increased – so the isotherms increase in  $T$  as one moves away from the  $V$ -axis.

### 1.3 Paths and functions of state

We have constructed isotherms in the  $p$ - $V$  plane, which are states with the same temperature. Can we describe the system moving along them? Yes as long as we change, say, the volume in a "quasi-static" fashion. This means it is slow compared to the *relaxation time* of the system, after which the pressure and temperature are uniform across the system. In Mathematics 2B

the *heat equation* will be solved. It describes the variation of the temperature as a function of time and space, in one dimension,  $T(x, t)$  in a medium obeys:

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2}$$

where the constant  $\kappa$  is the "thermal diffusivity". By dimensional analysis, if the system has size  $L$ , then the relaxation time is

$$\tau \simeq \frac{L^2}{\kappa} ,$$

so the relaxation time grows with system size (this will be explained more clearly in Mathematics 2B). Quasi-static would mean that the fractional change in temperature over the relaxation time is small compared to the initial temperature, i.e.

$$\frac{T(t + \tau) - T(t)}{T(t)} \ll 1 .$$

Simple classes of quasi-static paths are:

- *isothermal* processes, where the system moves along an isotherm;
- *adiabatic* processes, where the system is in an adiabatic container – there is no flow of heat;
- *isobaric* processes, where the pressure is constant;
- and *isochoric* processes, where the volume is constant, so no work is performed (by a piston)<sup>1</sup>.

A crucial important idea has slid into the discussion without its significance being observed. In the wake of the discussion of isotherms above, an unstated implication was that for any given point in the  $p$ - $V$  plane there is a unique associated temperature,  $T(p, V)$  and combining this with the discussion of paths, this would appear to be independent of path to  $(p, V)$  from, say,  $(p_0, V_0)$ .

Temperature,  $T(p, V)$ , is the first *function of state* we have encountered – its value being independent of path. A very important function of state is the internal energy,  $U(p, V)$ . It is called "internal" as it assumes that the gas does not have an average motion, so no macroscopic kinetic energy, and no change in potential – e.g. due to the centre of gravity of the gas changing height.

We can imagine checking the absence of path dependence by performing different types of work on the gas (not applying heat yet), as in Fig. (1.4), where the gas is in a piston and

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<sup>1</sup>But mechanical work could be performed by other means – for example by a falling weight driving Joule's paddle wheels.

may expand, performing work on the external atmosphere. But, in addition there are "Joule" paddle wheels driven by a falling weight which increase the energy of the gas due to viscosity. In the figure you can see the parts of the path with no volume change, associated with the work from paddle wheels and the other sections due to piston motion. The two paths from  $A$  to  $B$ , via either  $C$  or  $D$ , could, in principle, be checked to see the total work done was the same – showing  $U(p, V)$  is a function of state.

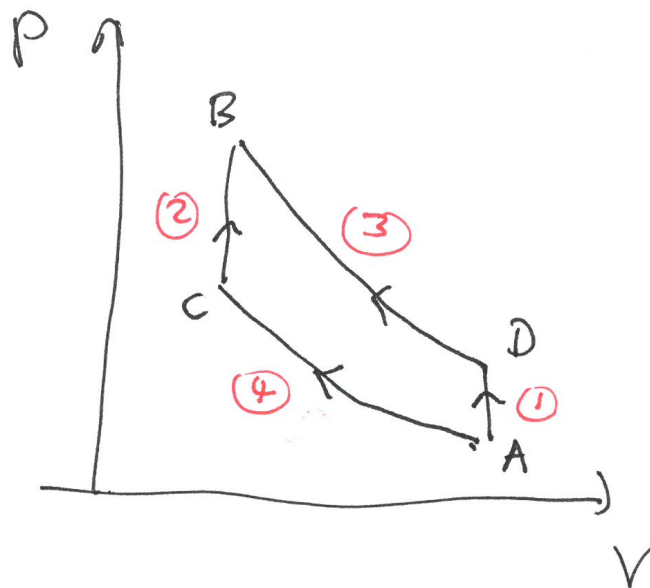


Figure 1.4: Two routes between  $A$  and  $B$  in the  $p$ - $V$  plane.

The variables one uses in a function of state can reflect the nature of the problem and also convenience. Below we will see an example where one chooses initially  $U(T, V)$  and then  $U(T, p)$ .

## 1.4 The First Law

In the wake of experiments by [Joule](#)<sup>2</sup> and others (see Y1 Temperature and Matter) it was realised that heat was a form of energy and the First Law was formulated to express overall conservation of energy:

$$\Delta U = Q_{\text{in}} + W_{\text{on}}$$

$$dU = dQ_{\text{in}} + dW_{\text{on}} .$$

<sup>2</sup>Joule was a brewer and his experiments were partly initially stimulated by an interest in making brewing more efficient. An account of the brewery side of his life is [here](#). there is a present-day brewery called [Joule's](#), which does not seem to be related to the original Joule, but with poetic licence one can experience Joule beer at the [New Inn](#) in Harborne (for example).

In the first equation,  $\Delta$  represents the change in a quantity,  $Q_{\text{in}}$  is the heat flowing *into* the system and  $W_{\text{on}}$  is the work performed *on* the system. It is very important to notice the signs associated with the heat and work.

The second equation expresses the First Law for small changes. Since  $U(p, V)$  is a function of state the total differential may be defined with respect to  $p$  and  $V$ :

$$dU = \left( \frac{\partial U}{\partial p} \right)_V dp + \left( \frac{\partial U}{\partial V} \right)_p dV . \quad (1.2)$$

where the subscripts on the partial derivatives state which variables are being held constant. However there is no sense (in general) to a system having a total amount of heat or work, so  $Q$  and  $W$  are NOT functions of state – paths between two points in the  $p$ - $V$  plane can be achieved with differing fractions of work and heat. Hence the special differential,  $\delta$ . We will return to this later.

An example of the utilisation of the First Law is deriving a relation between different specific heats. It will provide important practice in choosing and changing variables in a thermodynamic context. The problem being considered will always guide the choice of variables, for example if a variable (e.g.  $V$  or  $p$ ) is being kept fixed, make that one of the variables so the constancy is manifest.

One way to characterise heat is to write  $\delta Q = C dT$ ,  $C$  being a specific heat. If we are using a piston to perform work, then  $\delta W = -p dV$ . In this case the First Law becomes:

$$dU = C dT - p dV . \quad (1.3)$$

Note the right hand side implies, because it involves  $dT$  and  $dV$ , that we should regard  $U(T, V)$ , to be a function of  $T$  and  $V$ .

So let us write the total differential of  $U$  in terms of  $T$  and  $V$ :

$$dU = \left( \frac{\partial U}{\partial T} \right)_V dT + \left( \frac{\partial U}{\partial V} \right)_T dV . \quad (1.4)$$

Then we may identify the right hand sides of Eq. (1.2) and Eq. (1.4):

$$\left( \frac{\partial U}{\partial T} \right)_V dT + \left( \frac{\partial U}{\partial V} \right)_T dV = C dT - p dV .$$

Then if we chose a change at constant volume,  $dV = 0$ , so we find noting we have  $C_V$  as volume is fixed,:

$$\begin{aligned} \left( \frac{\partial U}{\partial T} \right)_V dT + \left( \frac{\partial U}{\partial V} \right)_T \underbrace{dV}_{=0} &= C_V dT - p \underbrace{dV}_{=0} \\ \Rightarrow \left( \frac{\partial U}{\partial T} \right)_V &= C_V . \end{aligned}$$

How do we obtain the specific heat at constant *pressure*,  $C_p$ ? The variables should be  $T$  and  $p$ , so we can set  $dp = 0$ . Thus we must write the total differentials  $dU$  and  $dV$  in terms of  $T$  and  $p$ :

$$\begin{aligned}
 dU &= \left(\frac{\partial U}{\partial T}\right)_p dT + \left(\frac{\partial U}{\partial p}\right)_T dp = C dT - p \underbrace{\left[\left(\frac{\partial V}{\partial T}\right)_p dT + \left(\frac{\partial V}{\partial p}\right)_T dp\right]}_{=dV} \\
 \text{(dp = 0)} \quad \left(\frac{\partial U}{\partial T}\right)_p dT &= C_p dT - p \left(\frac{\partial V}{\partial T}\right)_p dT \\
 \Rightarrow C_p &= \left(\frac{\partial U}{\partial T}\right)_p + p \left(\frac{\partial V}{\partial T}\right)_p .
 \end{aligned} \tag{1.5}$$

We will see how this produces the usual result for an ideal gas below; *here* we have allowed a general equation of state.

## 1.5 The ideal gas

Thermodynamics does not provide the concrete form for either the equation of state or the functional form of the internal energy. These come from experiment. Most gases at STP (pressure of one atmosphere,  $\simeq 10^5$  Pa, and  $0^\circ$  C) are ideal gases. Water is not – it is about to freeze. The ideal gas equation of state is

$$pV = nRT = Nk_B T$$

where  $n$  is the number of moles,  $R \simeq 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$  is the gas constant,  $N$  is the total number of gas particles,  $k_B = R/N_A \simeq 1.381 \times 10^{-23} \text{ J K}^{-1}$  is Boltzmann's constant and  $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$  is Avogadro's number.

The experimental evidence for most atmospheric gases being ideal is depicted in Fig. (1.5) where, as the pressure vanishes,  $p \rightarrow 0$ , one sees the ideal gas equation of state is obeyed.

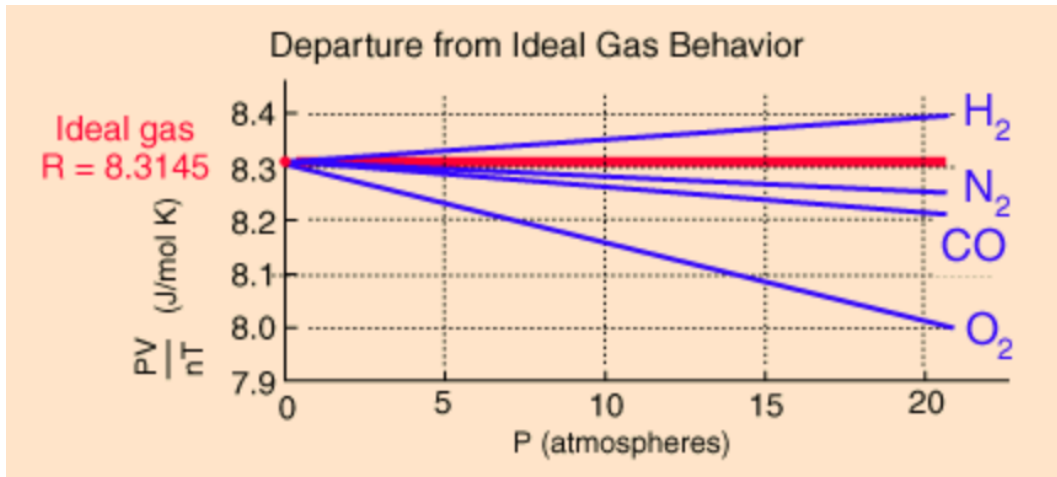
The other ingredient from experiment is that the internal energy,  $U(T)$ , is only a function of  $T$  and not, for example  $V$ . This was shown by Joule's "second experiment". What function of  $T$ ? That depends on the system. For example for atmospheric pressure hydrogen gas,  $C_v$  is shown in Fig. (1.6). Notice that for an ideal gas,  $U(T)$  only:

$$\left(\frac{\partial U}{\partial T}\right)_v = C_v = \frac{dU}{dT} = \left(\frac{\partial U}{\partial T}\right)_p .$$

From Fig. (1.6) we see that, depending on the temperature,  $c_v = mR$ , with  $m$  and integer reflecting how many degrees of freedom relevant (just centre of mass in 3d, or rotational and vibrational in addition). Notice the specific heat is almost a set of constant values as a function of  $T$ . For the lowest region ( $T \lesssim 100\text{K}$ ), for example, we can say that

$$U = U_0 + \frac{3}{2}RT$$





**Figure 1.5:** The approach to an ideal gas as  $p \rightarrow 0$  – notice where atmospheric pressure resides on the horizontal axis.

integrating  $c_v = (3/2)R$  with respect to  $T$ , where  $U_0 = U(T = 0)$ .

We may now substitute this information (equation of state and specific heat) into Eq. (1.5) to see

$$\begin{aligned}
 C_p &= \left( \frac{\partial U}{\partial T} \right)_p + p \left( \frac{\partial V}{\partial T} \right)_p \\
 \text{(Use eqn. state.)} \quad &= C_v + p \left( \frac{\partial}{\partial T} \frac{nRT}{p} \right)_p \\
 &= C_v + p \frac{nR}{p} \\
 \Rightarrow \quad C_p &= C_v + nR .
 \end{aligned}$$

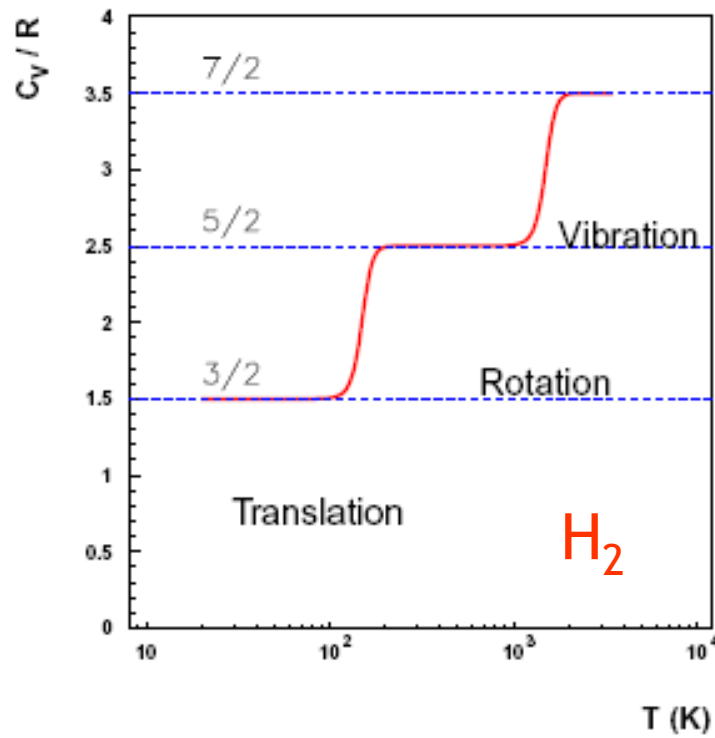
A familiar result from Temperature and Matter.

## 1.6 Adiabatic processes for ideal gases

For adiabatic processes,  $\delta Q = 0$ . We seek a relation between  $p$  and  $V$  describing an adiabatic process in the  $p$ - $V$  plane. This another exercise in changing variables in total derivatives.

The First Law becomes, upon expressing the total derivatives in terms of  $p$  and  $V$ , and utilising  $U$  being only a function of  $T$

$$\begin{aligned}
 dU &= \frac{dU}{dT} dT = -pdV \\
 \Rightarrow \quad C_v \left[ \left( \frac{\partial T}{\partial p} \right)_v dp + \left( \frac{\partial T}{\partial V} \right)_p dV \right] &= -pdV , \tag{1.6}
 \end{aligned}$$



**Figure 1.6:** The specific heat at constant volume for hydrogen gas.

Now we use the equation of state,  $pV = nRT$ , to determine the partial derivatives in Eq. (1.6):

$$\left(\frac{\partial T}{\partial p}\right)_V = \left(\frac{\partial}{\partial p} \frac{pV}{nR}\right)_V = \frac{V}{nR} \quad (1.7)$$

$$\left(\frac{\partial T}{\partial V}\right)_p = \left(\frac{\partial}{\partial V} \frac{pV}{nR}\right)_p = \frac{p}{nR} \quad (1.8)$$

Substituting Eq. (1.7) and Eq. (1.8) into Eq. (1.6) we see

$$\begin{aligned} \frac{C_V}{nR}(Vdp + pdV) &= -p \frac{nR}{nR} dV \\ \Rightarrow C_V V dp &= -pdV (C_V + nR) \\ \Rightarrow \frac{dp}{p} &= -\frac{C_V + nR}{C_V} \frac{dV}{V} . \end{aligned}$$

Then noting  $C_p = C_V + nR$  and denoting  $C_p/C_V = \gamma$ , we see upon integration

$$\begin{aligned} \ln p &= -\gamma \ln V + \ln D \quad (\ln D \text{ for convenience.}) \\ \Rightarrow \ln p + \gamma \ln V &= \ln D \\ \Rightarrow pV^\gamma &= D \quad (\text{A constant}) , \end{aligned} \quad (1.9)$$

as expected.

## 1.7 The Carnot cycle

A particularly simple cyclic process is central to the subject of the next section, the Second Law of Thermodynamics. This is the *Carnot cycle*. The Carnot cycle in Fig. (1.7) consists of two

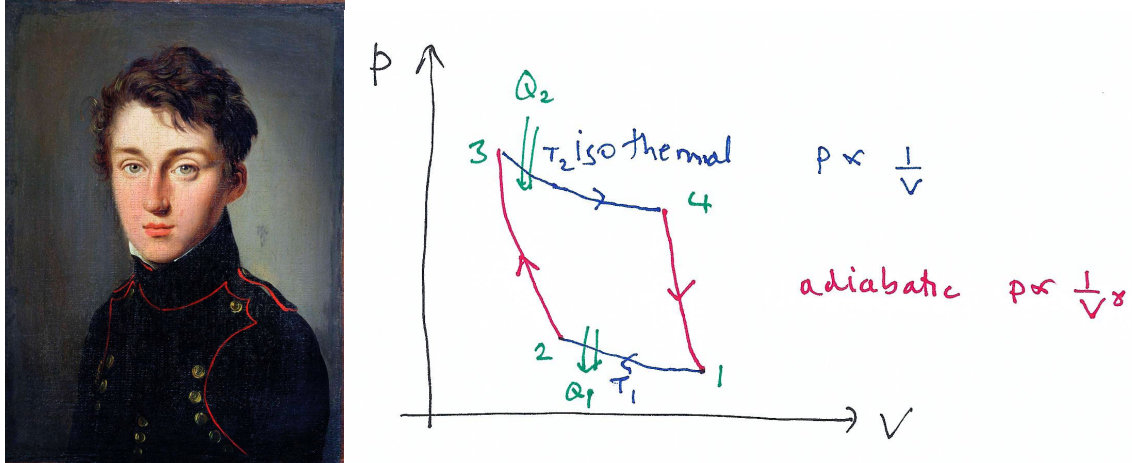


Figure 1.7: Left: Sadi Carnot. Right: the Carnot cycle.

isothermal processes at  $T_1$  and  $T_2$  with associated heat output  $Q_1$  and input  $Q_2$  respectively. These are linked by two adiabatic processes. We will use an ideal gas as the fluid. In fact the results are more general, but in reality the fluid is often ideal, and the result is obtained more simply.

We will now establish the result for a Carnot cycle that

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2} \quad (1.10)$$

which presages the definition of entropy.

For an adiabatic process,  $dQ = 0$  and so  $dU = dW$ . For an ideal gas,  $dU = C_v dT$ , so integrating,

$$W^{2 \rightarrow 3} = C_v(T_2 - T_1) \quad \text{and} \quad W^{4 \rightarrow 1} = C_v(T_1 - T_2).$$

Note the work done is equal and opposite, so the work contributed by the adiabatic sections of the cycle is in total zero.

Using  $pV^\gamma$  is a constant along with the equation of state to eliminate  $p$  in preference for  $T$  we see that  $TV^{\gamma-1}$  is constant. Let us apply this to the two adiabatic legs:

$$T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1} \quad \text{and} \quad (1.11)$$

$$T_1 V_1^{\gamma-1} = T_2 V_4^{\gamma-1} \quad (1.12)$$

$$\begin{aligned} & \text{(Dividing Eq. (1.11) by Eq. (1.12).)} \quad \frac{V_2^{\gamma-1}}{V_1^{\gamma-1}} = \frac{V_3^{\gamma-1}}{V_4^{\gamma-1}} \\ \Rightarrow \quad & \frac{V_2}{V_1} = \frac{V_3}{V_4}. \end{aligned} \quad (1.13)$$

Along the isotherms the change in  $U$  is zero (ideal gas), So the heat in/output,  $Q_2$  and  $-Q_1$ , is related to the work performed:

$$Q_1 = -Q^{1 \rightarrow 2} = +W^{1 \rightarrow 2} \quad \text{and} \quad Q_2 = Q^{3 \rightarrow 4} = -W^{3 \rightarrow 4} . \quad (1.14)$$

We may use the equation of state to derive the work performed,  $-\int p \, dV$ ,

$$W^{1 \rightarrow 2} = - \int_{V_1}^{V_2} p \, dV = -nRT_1 \int_{V_1}^{V_2} \frac{dV}{V} = -nRT_1 \ln \left( \frac{V_2}{V_1} \right) \quad (1.15)$$

$$\text{(Similarly:)} \quad W^{3 \rightarrow 4} = -nRT_2 \ln \left( \frac{V_4}{V_3} \right) . \quad (1.16)$$

Substituting Eq. (1.15) and Eq. (1.16) into Eq. (1.14) we see

$$\begin{aligned} \frac{Q_1}{Q_2} &= \frac{T_1 \ln \left( \frac{V_2}{V_1} \right)}{T_2 \ln \left( \frac{V_3}{V_4} \right)} \\ \text{(Use Eq. (1.13).)} \quad \Rightarrow \quad \frac{Q_1}{Q_2} &= \frac{T_1}{T_2} \\ \Rightarrow \quad \frac{Q_1}{T_1} &= \frac{Q_2}{T_2} . \end{aligned} \quad (1.17)$$

This result is the building block of the Second Law, and is a nascent form of entropy conservation.

# Appendices

## A Equation of state from Zeroth Law (Non-examinable)

Let us consider two fluid systems,  $A$  and  $C$ , initially in equilibrium by themselves and now form a diathermal wall between them so they may exchange energy and come to a mutual equilibrium.

We cannot choose all of  $p_A$ ,  $V_A$ ,  $p_C$  and  $V_C$  as independent (we could choose them all independently before the diathermal contact was created). Thus there is an equation

$$F_{A,C}(p_A, V_A; p_C, V_C) = 0$$

which expresses this lack of independence. It may be solved for  $p_C$  to yield

$$p_C = f_{A,C}(p_A, V_A; V_C) . \quad (\text{A.1})$$

Now consider *three* systems,  $A$ ,  $B$  and  $C$  all in equilibrium. The equilibrium between  $B$  and  $C$  provides

$$F_{B,C}(p_B, V_B; p_C, V_C) = 0 .$$

This may also be solved for  $p_C$  to yield

$$p_C = f_{B,C}(p_B, V_B; V_C) . \quad (\text{A.2})$$

We may now equate the two expressions for  $p_C$ , Eq. (A.1) and Eq. (A.2), to obtain:

$$f_{A,C}(p_A, V_A; V_C) = f_{B,C}(p_B, V_B; V_C) . \quad (\text{A.3})$$

But since we have another equation expressing that  $A$  and  $B$  are in equilibrium:

$$F_{A,B}(p_A, V_A; p_B, V_B) = 0 . \quad (\text{A.4})$$

Now Eq. (A.3) contains  $V_C$ , while Eq. (A.4) does not. So the dependence on  $V_C$  on each side of Eq. (A.3) must cancel. For example

$$f_{A,C}(p_A, V_A; V_C) = \phi_1(p_A, V_A)\zeta(V_C) + \eta(V_C)$$

and similarly for  $f_{B,C}(p_B, V_B; V_C)$ .

Thus we obtain

$$\phi_1(p_A, V_A) = \phi_2(p_B, V_B) .$$

and clearly by repeating this for  $B$  and  $C$  we see

$$\phi_1(p_A, V_A) = \phi_2(p_B, V_B) = \phi_3(p_C, V_C) .$$

Since the variables on each side of the above equalities are independent, each function must equal a constant, say  $T$ , the temperature. Of course it could be a function of the temperature.

Thus we arrive at the equations of state of the three systems:

$$\phi_1(p_A, V_A) = T .$$

With the obvious counterparts for  $B$  and  $C$ .